

Smart Climate Adaptive Roofing System (SCARS)

Executive Technical Overview

Problem Statement –

A growing labour shortage has become a major barrier to agricultural productivity. With fewer young people entering the sector, farms struggle to maintain regular operations. Factors like changing climate and inefficient resource use further discourage workforce participation, intensifying the shortage and threatening the long-term sustainability of farming.

Proposed Solution –

To design and implement a **solar-powered, AI-IoT enabled adaptive roofing system** that automates microclimate regulation, irrigation, and crop protection, thereby **reducing manual labour dependency** while improving climate resilience and increasing operational efficiency in open-field farming.

Agricultural productivity is increasingly constrained by a severe and growing labour shortage. Key field activities—irrigation regulation, shade management, fertigation scheduling, and routine crop monitoring—traditionally rely on continuous manual intervention. With an aging agricultural workforce and declining youth participation, farms struggle to maintain timely crop care, leading to avoidable yield losses, higher resource wastage, and increased vulnerability to climatic and pest-related stresses.

SCARS (Smart Climate Adaptive Roofing System) provides a scalable, automated solution designed to minimize labour dependency while improving crop resilience and operational efficiency. The system integrates solar-powered adjustable roofing panels with embedded micro-irrigation and fertigation infrastructure, supplemented by an IoT-enabled environmental sensing network. SCARS regulates sunlight exposure, rainfall shielding, irrigation volume, and nutrient application through a microcontroller-based control unit capable of fully autonomous operation, even without constant internet connectivity—ensuring suitability for remote and rural agricultural environments.

By reducing manual workload and maintaining optimized crop microclimate conditions, SCARS enhances yield stability, resource efficiency, and long-term sustainability across diverse crop categories.

Technical Methodology & System Architecture

1. System Structure

SCARS functions as an integrated precision agriculture platform, combining:

- Mechanized roofing panels with servo-based angle control.
- Soil moisture, light intensity, rainfall, nutrient, and pest monitoring sensors.
- Embedded micro-irrigation and fertigation delivery pipelines.
- ESP32 C6 microcontroller unit with on-board logic for autonomous decision-making.
- Mobile/web interface for user oversight and manual operation.

All subsystems communicate centrally, ensuring synchronized climate, irrigation, and nutrient management.

2. Mode of Operation

2.1 Manual Mode

Allows direct user control for real-time field management:

- **Irrigation Control:** On-demand sprinkler activation using live soil moisture data.
- **Fertigation & Pesticide Application:** Controlled nutrient and pest treatment cycles through integrated pipelines.
- **Sunlight Management:**
 - Sun Tracking: Dynamic slab orientation using dual-LDR feedback.
 - Partial or Full Shade modes to mitigate heat stress.
- **Rainfall Regulation:** Selection between full exposure, moderated exposure, or complete crop protection.

2.2 Automatic Mode

Transforms SCARS into a fully autonomous microclimate system:

- User selects crop type (e.g., Wheat, Rice, Maize, Sugarcane).
- System applies preloaded crop-specific environmental thresholds.
- **Automated Irrigation:** Triggered by soil moisture deviations.

- **Smart Fertigation & Pest Response:** Adjusted based on nutrient sensor and pest monitoring inputs.
- **Dynamic Light and Rain Adaptation:** Panel angles regulated to maintain optimal photosynthetic and hydrological balance.

3. Emergency Priority — Hailstorm Defense

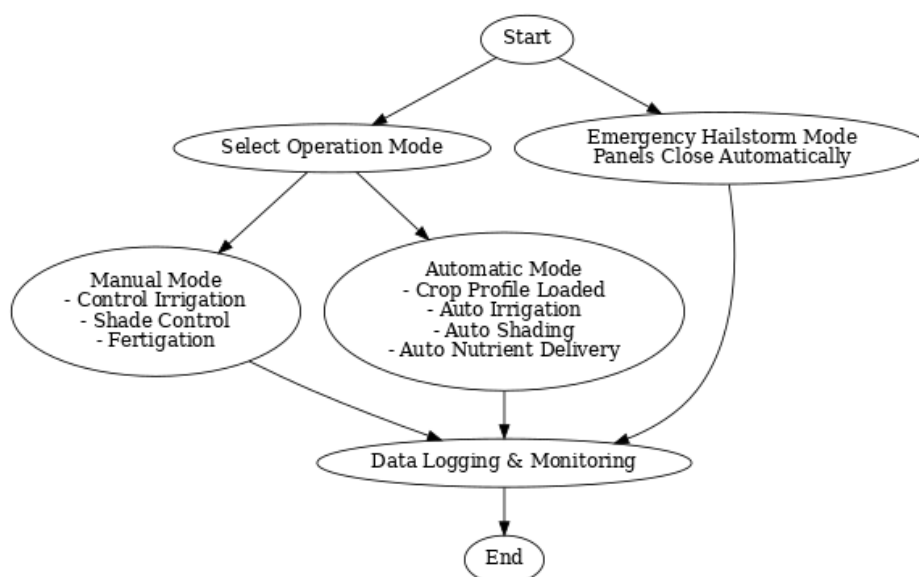
In extreme conditions, SCARS overrides all other operations:

- Roofing panels immediately close to shield crops.
- System automatically resumes normal function once conditions stabilize.

4. Data Processing & Connectivity

- All sensing and control computations are handled locally for uninterrupted offline functionality.
- Dashboard connectivity via Wi-Fi or LoRa enables remote monitoring when available.
- Real-time operational analytics are continuously logged for performance benchmarking and yield optimization.

5. Flowchart



5. Operational Flow Summary

1. User selects **Manual** or **Automatic** mode.
2. Manual mode provides direct control with live sensor feedback.
3. Automatic mode self-regulates climate, irrigation, and nutrient delivery per crop profile.
4. Emergency mode safeguards crops from hailstorm damage.
5. System logs operational and environmental data for analysis.

6. Block Diagram

